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2016-2017

M1 MACRO

Group B

Session II

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

$$\text{Max}_{A_{t+1}, A_{t+2}, \dots} E_t \left[u \left[(1+r)A_t + Y_t - A_{t+1} \right] + \beta u \left[(1+r)A_{t+1} + Y_{t+1} - A_{t+2} \right] + \dots \right]$$

FOC wrt A_{t+1} : $E_t \left[-u'(c_t) + \beta (1+r) u'(c_{t+1}) \right] = 0$

Euler equation $\nearrow$

$$\Leftrightarrow \underbrace{E_t [u'(c_t)]}_{u'(c_t)} = \underbrace{\beta (1+r)}_{=1} E_t [u'(c_{t+1})]$$

(BY ASSUMPTION)

STOCHASTIC EULER EQUATION:

$$u'(c_t) = E_t u'(c_{t+1})$$

ASSUMPTION

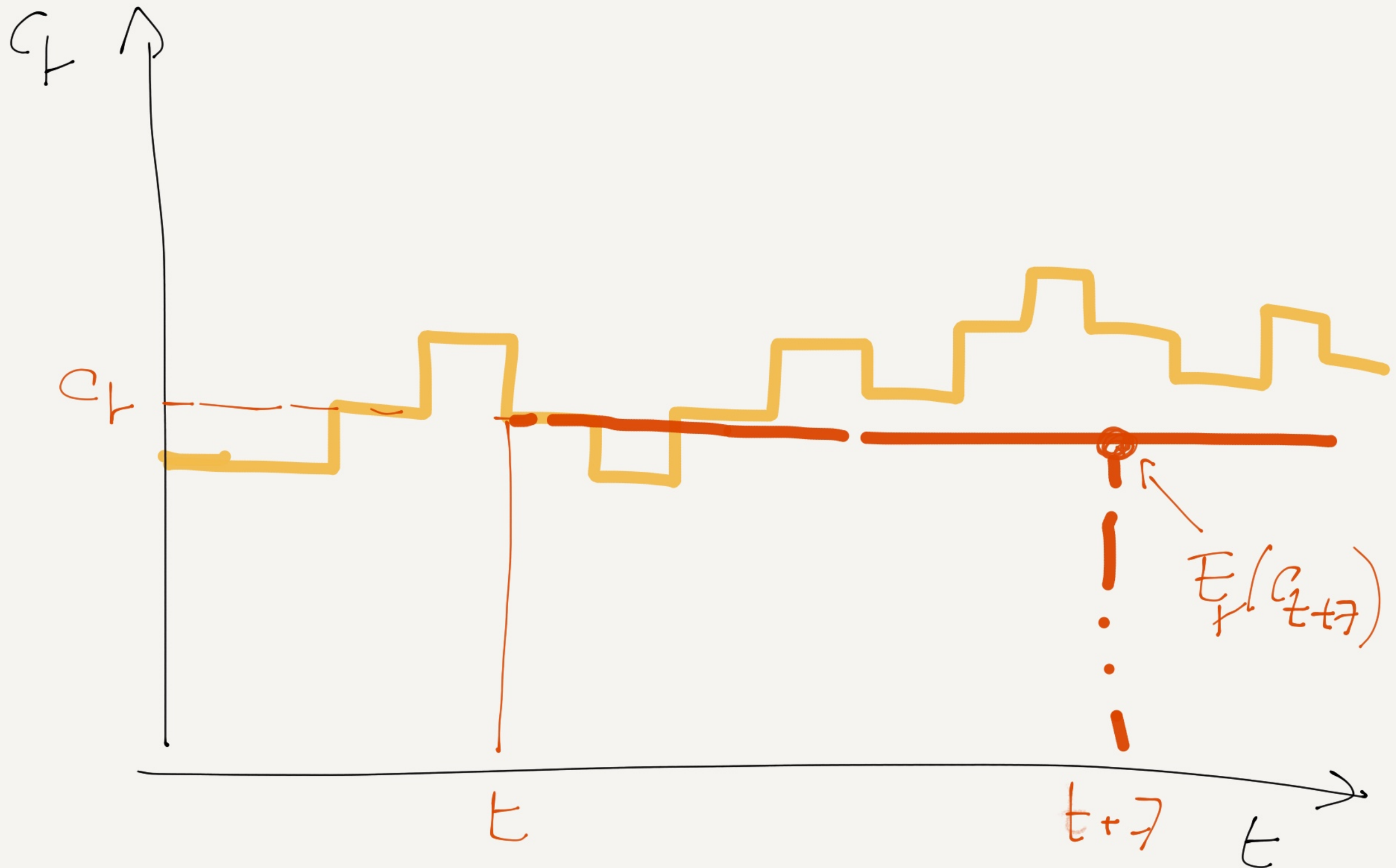
$$u(c_t) = \alpha_0 c_t + \frac{\alpha_1}{2} c_t^2$$

$$u'(c_t) = \alpha_0 + \alpha_1 c_t$$

$$\alpha_0 + \alpha_1 c_t = E_t (\alpha_0 + \alpha_1 c_{t+1})$$

$$c_t = E_t (c_{t+1})$$

CONSUMPTION FOLLOWS A RANDOM WALK



$$A_t = \frac{1}{1+r} A_{t+1} + Y_t - C_t$$

$$= \frac{1}{1+r} \left(\frac{1}{1+r} A_{t+2} + Y_{t+1} - C_{t+1} \right) + Y_t - C_t$$

$$= \left(\frac{1}{1+r} \right)^2 A_{t+2} + \frac{1}{1+r} (Y_{t+1} - C_{t+1}) + (Y_t - C_t)$$

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$$A_t = \left(\frac{1}{1+r} \right)^{T+1} A_{t+T+1} + \frac{1}{1+r} \sum_{i=0}^T \frac{Y_{t+i} - C_{t+i}}{(1+r)^i}$$

$$E_t A_t = E_t [\quad]$$

$$A_t = \lim_{T \rightarrow \infty} E_t \frac{A_{t+T+1}}{(1+r)^{T+1}} + \frac{1}{1+r} \sum_{i=0}^{\infty} \frac{E_t(Y_{t+i}) - E_t(C_{t+i})}{(1+r)^i}$$

$$= 0$$

TRANSVERSALITY
CONDITION

TOTAL (DISC.)
EXPENDITURES

↔

$$A_t + \underbrace{\frac{1}{1+r} \sum_{i=0}^{\infty} \frac{E_t Y_{t+i}}{(1+r)^i}}_{H_t} =$$

$$\frac{1}{1+r} \sum_{i=0}^{\infty} \frac{E_t(C_{t+i})}{(1+r)^i}$$

Euler eq.

$$E(c_{t+i}) = C_t \quad \forall i$$

Therefore,

$$\begin{aligned} A_t + H_t &= \frac{1}{1+r} \sum \frac{C_t}{(1+r)^i} \\ &= \frac{C_t}{1+r} \times \frac{1}{1 - \frac{1}{1+r}} = \frac{C_t}{r} \end{aligned}$$

Solution

↪

$$C_t = r (A_t + H_t)$$

SPECIFY A PROCESS FOR y_t

$$W_t = \frac{1}{W_2} \sum_{i=0}^{\infty} \frac{E_t y_{t+i}}{(1+r)^i}$$

$$y_t = \bar{y} + \tilde{y}_t$$

↑ permanent component of income

↑ temporary component of income

ASSUME

$$\tilde{y}_t = \rho \tilde{y}_{t-1} + \varepsilon_t, \quad 0 \leq \rho \leq 1$$

→ AR(1) process

Simple case $\boxed{\rho=0}$: $\tilde{y}_t = \varepsilon_t$

$\rightarrow E_t \tilde{y}_{t+i} = E_t \varepsilon_{t+i} = 0$

$$H_t = \frac{1}{1+n} \sum_{i=0}^{\infty} \frac{E_t (\bar{y} + \tilde{y}_{t+i})}{(1+n)^i}$$

$$= \underbrace{\frac{1}{1+n} \sum_{i=0}^{\infty} \frac{\bar{y}}{(1+n)^i}}_{\frac{\bar{y}}{n}} + \frac{1}{1+n} \left[\underbrace{E_t \tilde{y}_t}_{\varepsilon_t} + \underbrace{\sum_{i=1}^{\infty} \frac{E_t \tilde{y}_{t+i}}{(1+n)^i}}_0 \right]$$

$$= \bar{y}/n + \varepsilon_t/(1+n)$$

then,

$$C_t = r A_t + r \left(\frac{\bar{y}}{r} + \frac{\varepsilon_t}{1+r} \right)$$

$$C_t = r A_t + \bar{y} + \frac{r}{1+r} \varepsilon_t$$

→ permanent component

→ temporary component

marginal propensity to consume

- permanent income $\frac{dC_t}{d\bar{y}} = 1$

- temp income: $\frac{dC_t}{d\varepsilon_t} = \frac{r}{1+r}$

Now look at the case $\tilde{y}_t = \rho \tilde{y}_{t-1} + \varepsilon_t$

$$\begin{aligned} \hookrightarrow E_f(\tilde{y}_{t+1}) &= E_f(\rho \tilde{y}_t + \varepsilon_{t+1}) = \rho E_f(\tilde{y}_t) + E_f(\varepsilon_{t+1}) \\ &= \rho \tilde{y}_t \end{aligned}$$

$$\boxed{E_f[\tilde{y}_{t+i}] = \rho^i \tilde{y}_t}$$

Then
$$h_t = \frac{\bar{y}}{n} + \frac{1}{1+\alpha} \sum_{i=0}^{\infty} \frac{E_f \tilde{y}_{t+i}}{(1+\alpha)^i}$$

$$= \frac{\bar{y}}{n} + \frac{1}{1+\alpha} \tilde{y}_t \times \sum_{i=0}^{\infty} \frac{\rho^i}{(1+\alpha)^i} = \frac{\bar{y}}{n} + \frac{\tilde{y}_t}{1+\alpha} \times \frac{1}{1 - \frac{\rho}{1+\alpha}}$$

$$H_t = \frac{\bar{y}}{n} + \frac{1}{1+n} \tilde{y}_t \quad \frac{h^2}{1+n-\rho}$$

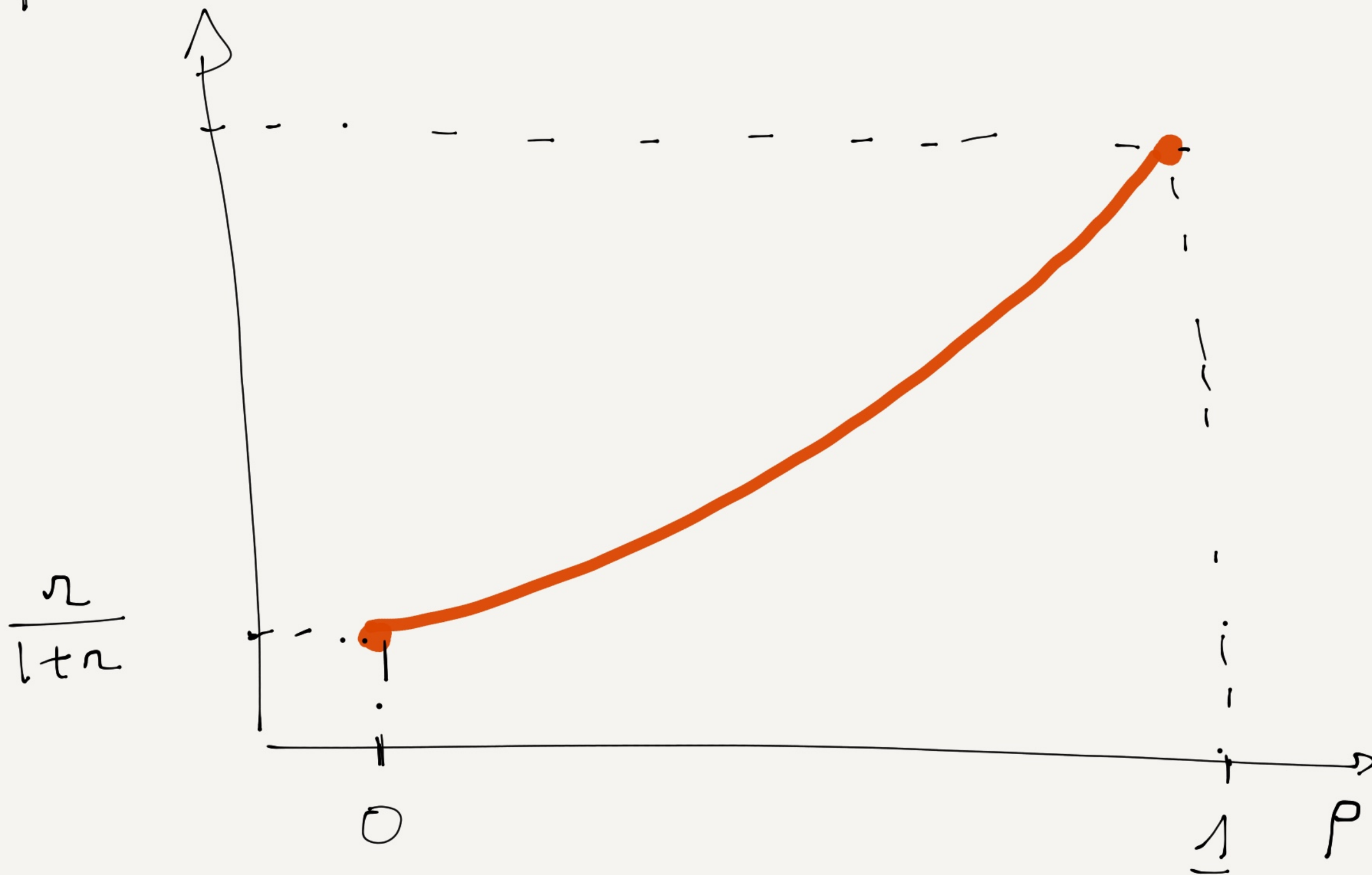
$$= \frac{\bar{y}}{n} + \frac{\tilde{y}_t}{4n-\rho}$$

$$\hookrightarrow C_t = n A_t + \underbrace{\bar{y}}_{\text{permanent}} + \frac{n}{1+n-\rho} \underbrace{\tilde{y}_t}_{\text{temporary, persistent}} \text{ but}$$

$$\frac{dC_t}{d\bar{y}} = 1, \quad \frac{dC_t}{d\tilde{y}_t} = \frac{n}{1+n-\rho}$$

$$\frac{1}{-n}$$

$m p <$



Note

$$\begin{aligned} E f(x) &\neq f(E(x)) \\ &\left(= \text{if } f \text{ is linear} \right) \end{aligned}$$